EXPERIMENT 13

Parse Tree Generation Using Context-Free Grammar

Aim

To generate and display a parse tree for a sentence using a context-free grammar.

Procedure

1. Import CFG and RecursiveDescentParser from NLTK.
2. Define the grammar rules.
3. Provide the input sentence as tokens.
4. Generate the parse tree.
5. Print the tree representation.

Program

```
# Install first: pip install nltk
from nltk import CFG
from nltk.parse import RecursiveDescentParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V
Det -> 'the'
N -> 'student'
V -> 'reads'
""")
parser = RecursiveDescentParser(grammar)
for tree in parser.parse("the student reads".split()):
    print(tree)
    tree.pretty_print()
```

Output

```
OUTPUT — EXPERIMENT 13
(S (NP (Det the) (N student)) (VP (V reads)))
A tree diagram is printed in the terminal.
```

Result

Thus, a parse tree was generated for the sentence using the given grammar.